Title: Next-Gen WAF - Cloud Engine & Deployment Options
Course Details:
Fastly's Next-Gen WAF is powered by Fastly's Signal Sciences platform. In this course we'll explain the Cloud Engine's role along with the various Deployment Options available for where your Agent/Module pairs can be deployed.
Learning Objectives:

- What are the Plan Tiers, and what do they gate or limit?
- What is the Signal Sciences Cloud Engine?
- Deployment Options
    - Edge Deployment
    - Cloud WAF Deployment
    - Core Deployment
    - Agent-Only Proxy Mode
    - Kubernetes
    - Envoy Proxy Integration

Lesson One: What are the Plan Tiers?
As of Q3 2022, there are three Platform Plan Tiers for using Next-Gen WAF. They are Essential, Professional, and Premier. As you step up from Essential to Professional and then to Premier, your Signal Sciences Corp will gain access to more features. Included below is a link out to our docs outlining which features are included in each Tier. As you progress through the other Next-Gen WAF courses, when we introduce a new feature we will call out which Tiers include that feature.

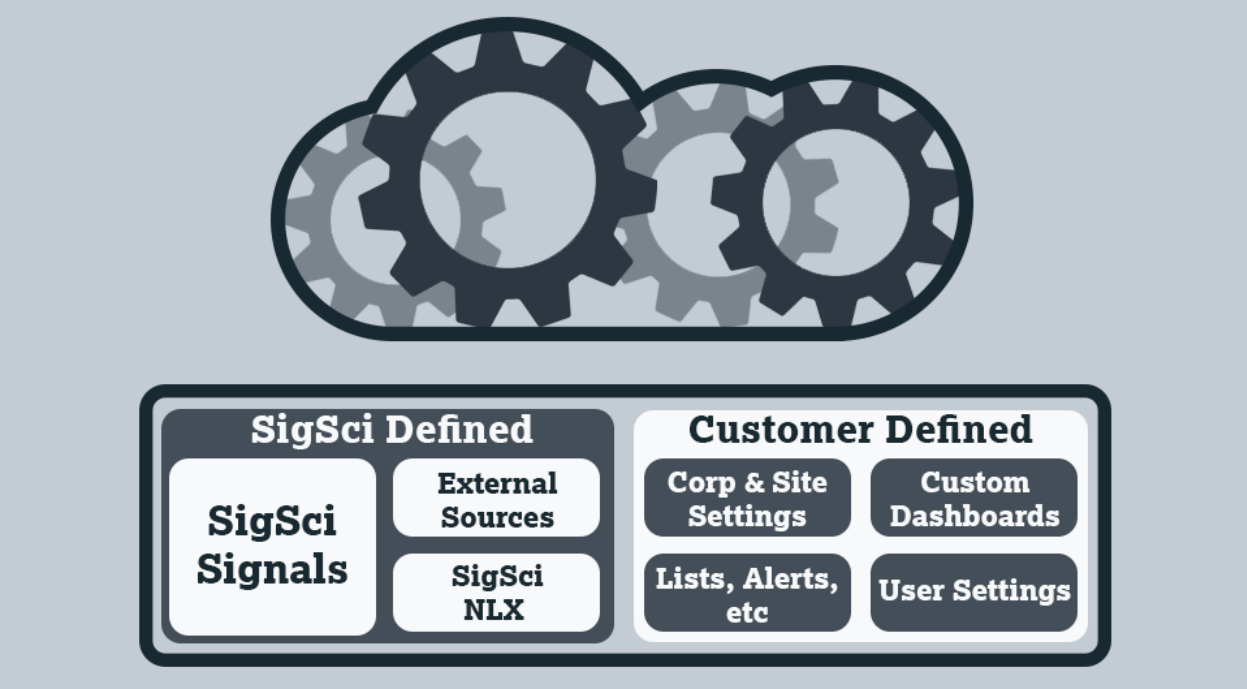
For features that are not included in your current Tier level, they will not be included in the UI nor be made available via API calls.

Next-Gen WAF Feature Availability

Relevant Links:

Lesson Two: The SigSci Cloud Engine

What is SigSci's Cloud Engine?



Signal Science's Cloud Engine is the cloud-hosted collection and analysis system that receives data from the agent(s) and other external sources. This includes request data, IP address information, and agent/module performance metrics. This information is then exported and made visible in the Signal Sciences console, through the API, and any third-party integrations you have set up.

An example of an external source used by the Cloud Engine is the TOR exit node feed. This data informs when the TORNODE signal is tagged onto a request originating from one of those TOR exit nodes. Same with the SANS signal, or Malicious IP Traffic, which is informed by the SANS Internet Storm Center.

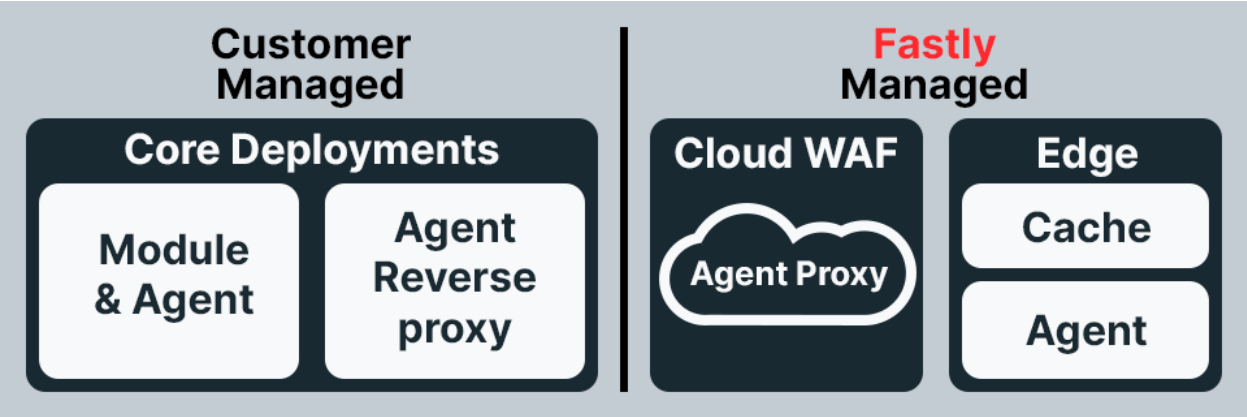
Additionally, internal monitoring of malicious IPs also informs our SIGSCI-IP, or Network Effect, signal. For the Network Effect signal, whenever an IP is flagged for crossing default Site Alert threshold by our decision engine, that IP will be propagated to all customers. We then log subsequent requests from those IPs that contain any additional signal for the duration of the flag. This is Signal Sciences (NLX) Network Learning Exchange in action.

The Cloud Engine is also the control plane for all the Customer defined settings, like Rules, Lists, Alerts, and many more which we'll cover in detail in later courses. Some of those settings, like Rules, will be propagated from the Cloud Engine to your deployed Agents, informing that Agent on which requests to Allow, Block, and/or Log.

Relevant Links:

https://www.signalsciences.com/blog/product-launch-2018-power-rules-and-network-learning-exchange/

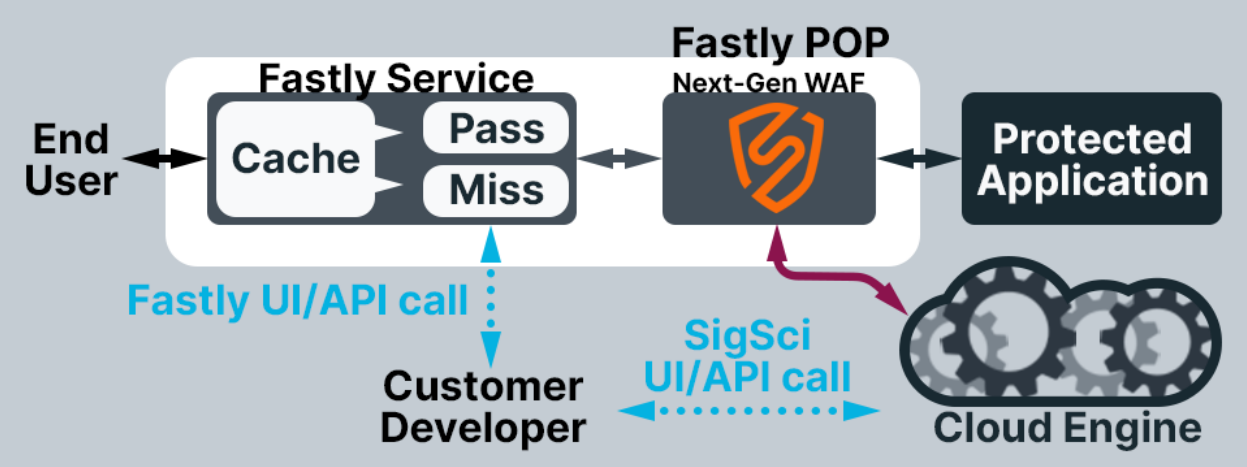
Lesson Three: Deployment Options
Signal Science has a variety of options available for where and how you want to integrate our Modules and Agents into your request flow. We can first categorize those options by Customer Managed & Fastly Managed.



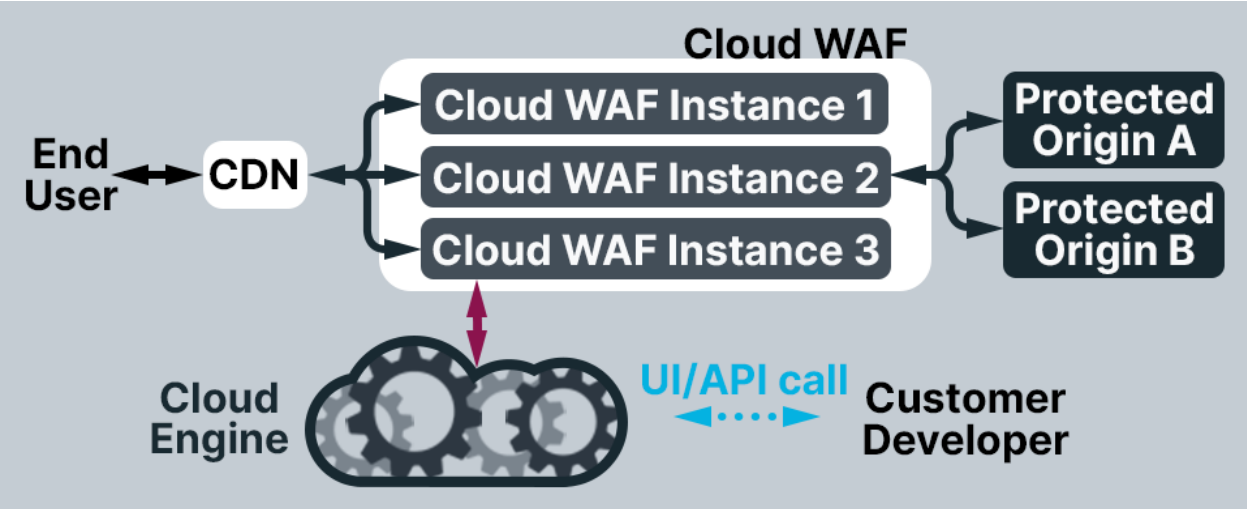
The **Customer Managed** options covers our Core Deployments, which allow our customers to install either an Module & Agent pair, or an Agent in Rever proxy mode within their infrastructure. After installed, customers are responsible for managing those deployments by pushing version updates as they are available.

Alternatively, customers can opt for either of our two **Fastly Managed** deployment options, Cloud WAF or Edge Deployment. For both of these options, Fastly will handle any version updates for any modules and/or agents. These two options also alleviate the need for customer managed infrastructure for our security solution to be installed on. Fastly manages that as well.

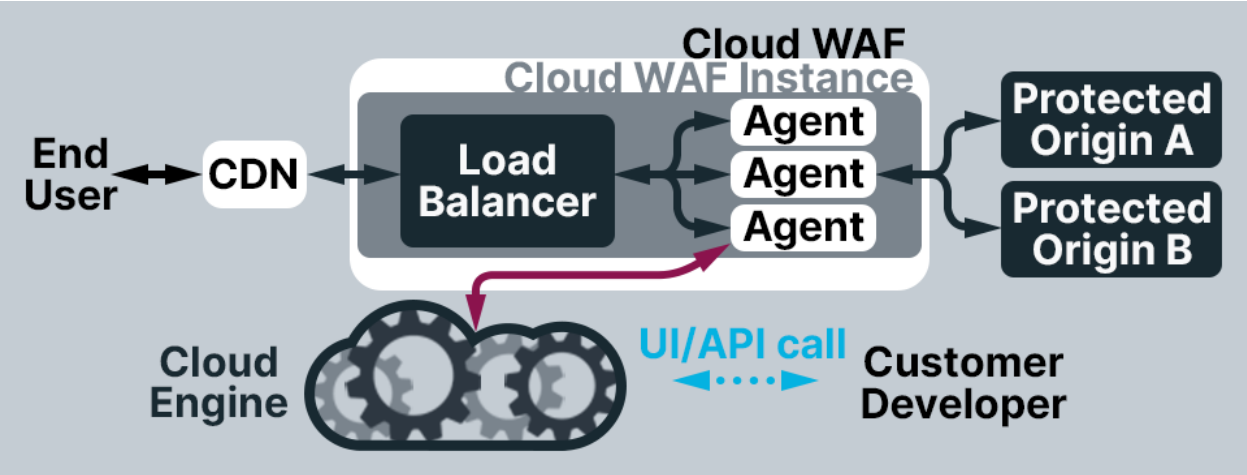
Let’s explore each of these options in more detail below, starting with Edge Deployment.



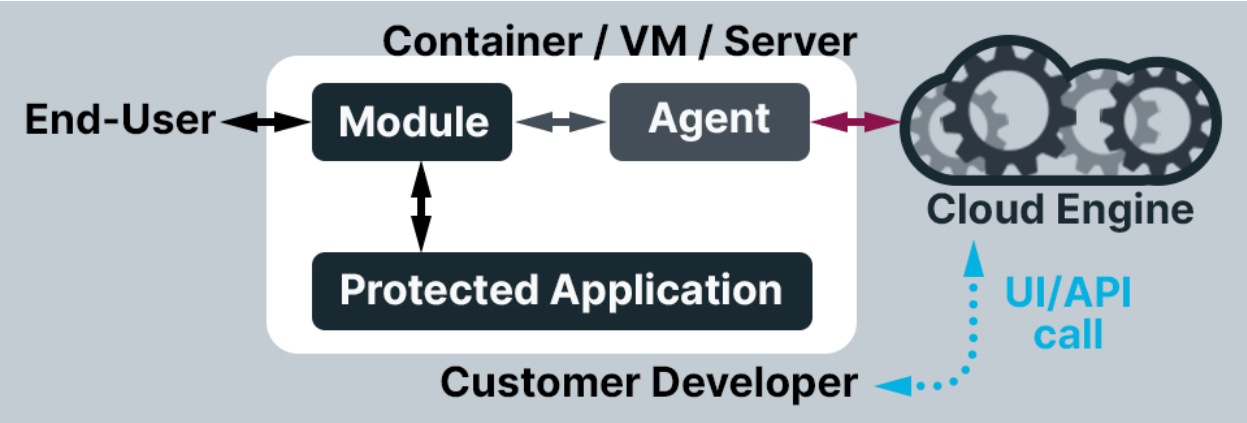
Edge Deployment: This option is hosted on Fastly's Edge Cloud platform, i.e. our global network of POPs. To use the edge deployment method, you must add it to new or existing Fastly services that you create in the Fastly console and update your DNS records to point to Fastly.



Cloud WAF Deployment: This option is hosted on Fastly's cloud-hosted infrastructure. To use Cloud WAF, you must upload a TLS certificate (or use Fastly for your certificate configuration), add an origin server using the Signal Sciences Hosted Dashboard, and update your DNS records to point to the appropriate servers. You can select to have Cloud WAF Instances configured in regions around the globe. What is a Cloud WAF Instance?

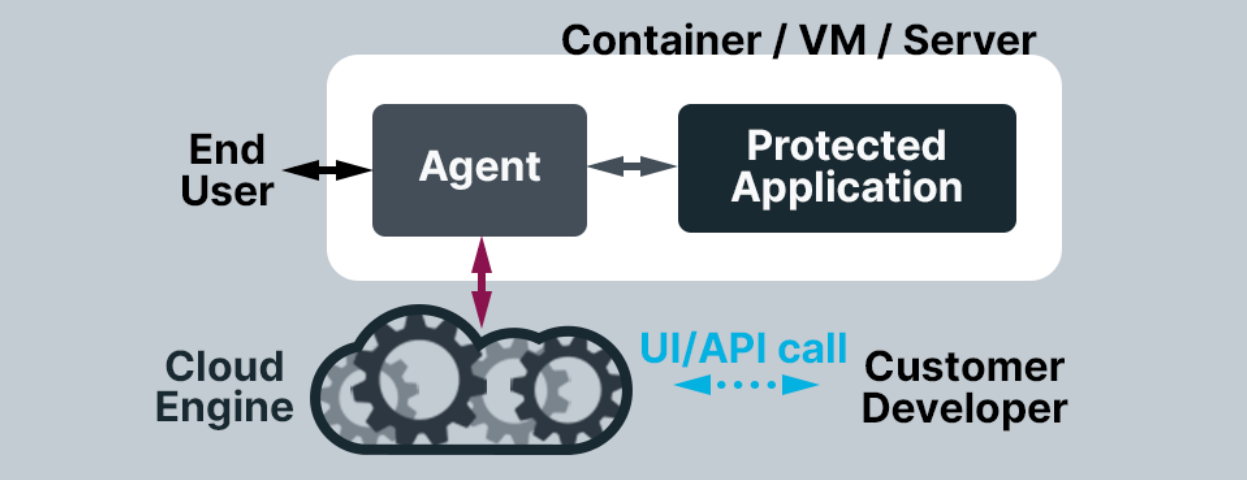


A **Cloud WAF Instance** is made up of a Load Balancer along with a minimum of three SigSci Agents, operating in Reverse Proxy mode. Each Agent is installed on separate redundant machines, and more Agents can be added to the Instance to meet traffic demands.



Core Deployment: Hosted directly on your web servers within your infrastructure. The Core deployment method consists of two components, the module and the agent. The module can exist as a plugin to your web server or as a language or framework specific implementation. The agent is a small daemon process that provides an interface between your web server and our cloud engine. You can also use the core deployment method without a module by running the agent in reverse proxy mode.

The Core Deployment is the most flexible of the three options. Depending on the particular platform and need, how the Agent and Module are installed and operate can vary. The first example of this is the Reverse Agent Proxy mode.



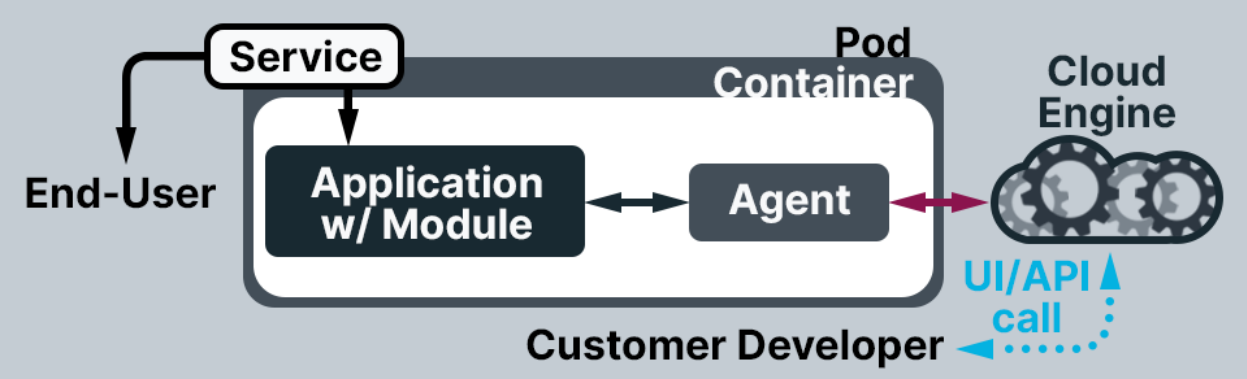
When an Agent is installed in Proxy Mode, it consolidates into itself the functions of the Module, actively Allowing or Blocking requests, while acting as a Proxy to your protected application.

Relevant Links:

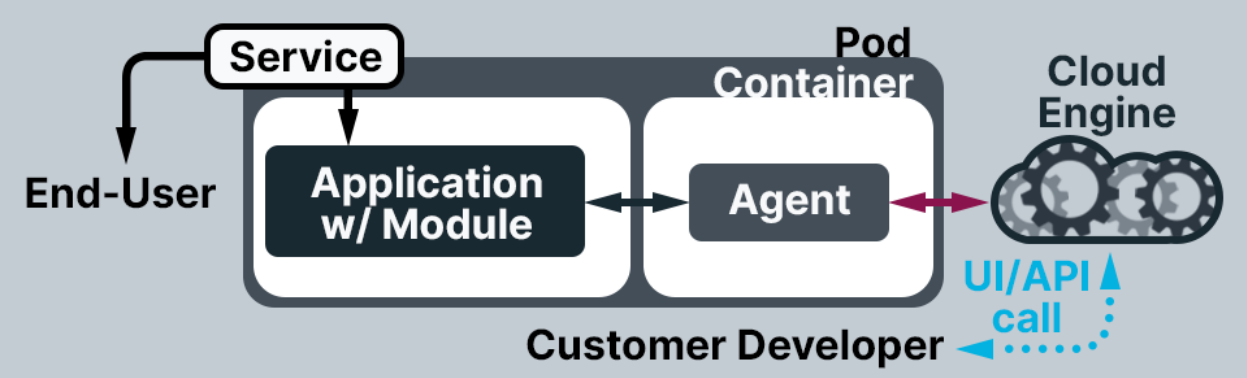
https://developer.fastly.com/learning/concepts/pop/

Lesson 4: Variations - Kubernetes Options

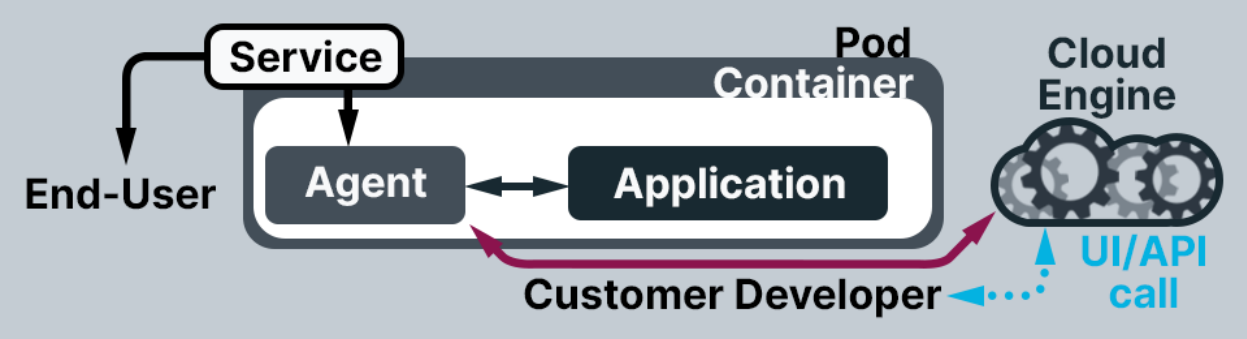
Installing Next-Gen WAF (Signal Science) into a Kubernees environment further demonstrates the variety of architecture options available. Multi-Pod vs Single Pod, Multi-Container vs Single, and Module/Agent vs Agent Proxy Mode, and you can understand how we end up with all the Deployment Options below.



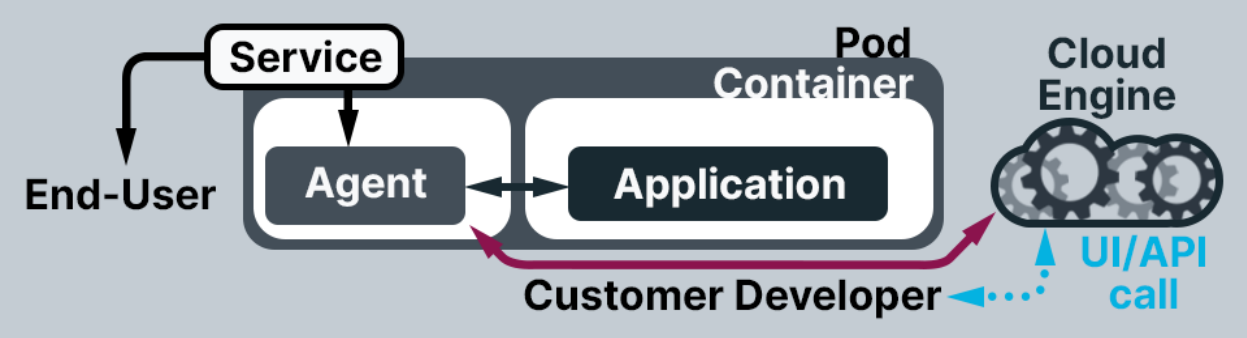
Single Pod, Single Container, Module and Agent Installation. The Module is installed in line with the protected application itself. The Agent is passed request data from the Module, and passes back a directive to Allow or Block the request.



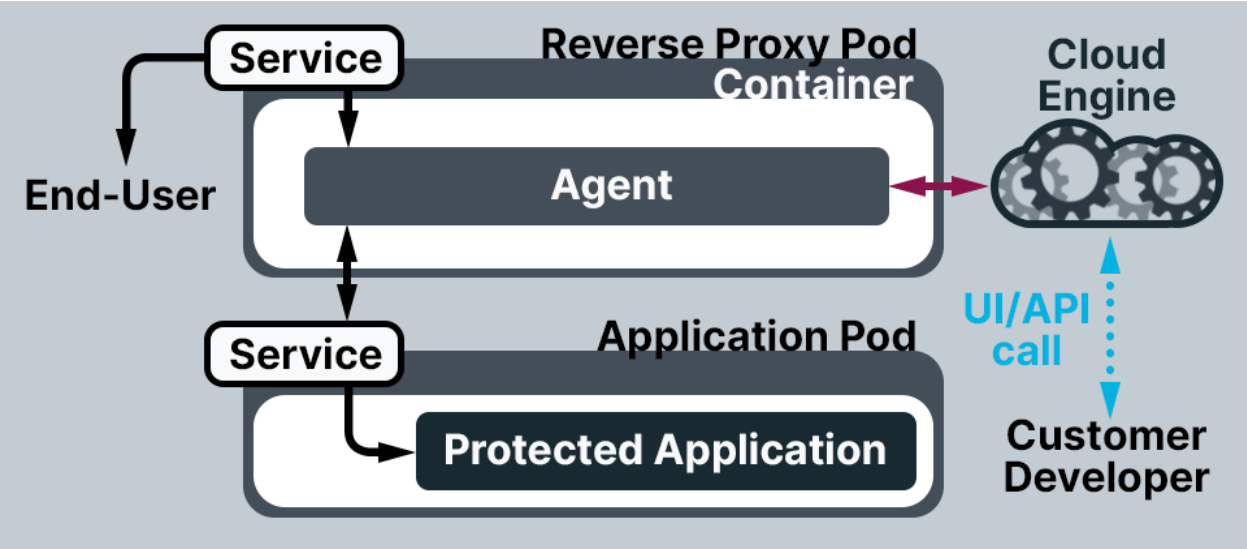
Single Pod, Multi-Container, Module and Agent. As needed, the Agent can be installed and configured to operate in a separate container from the Module.



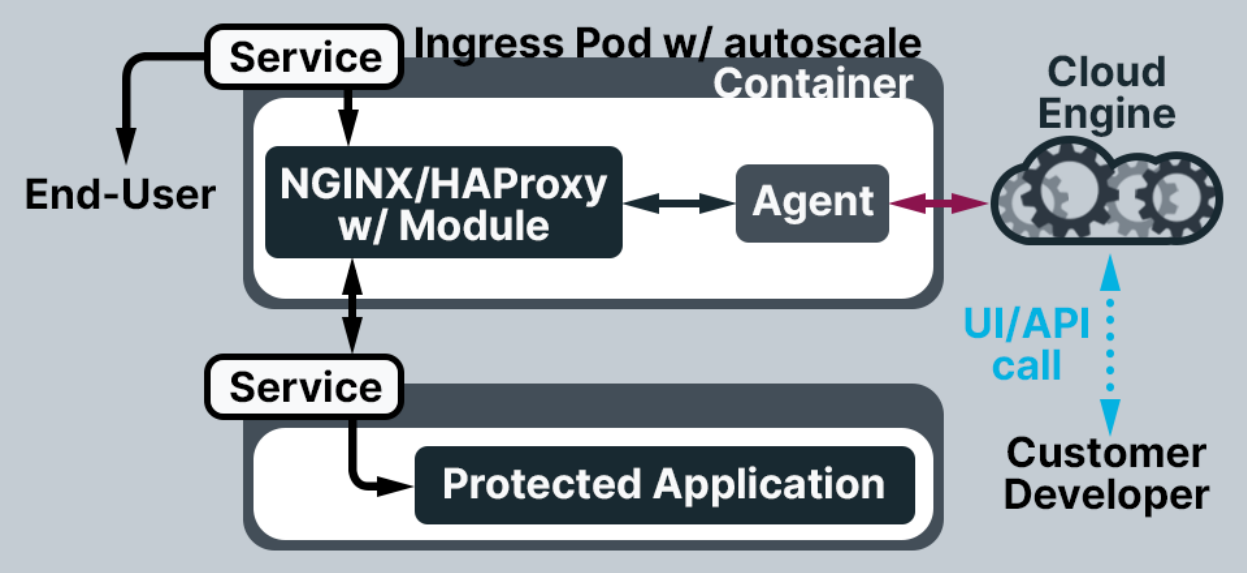
Single Pod, Single Container, Agent in Reverse Proxy Mode.



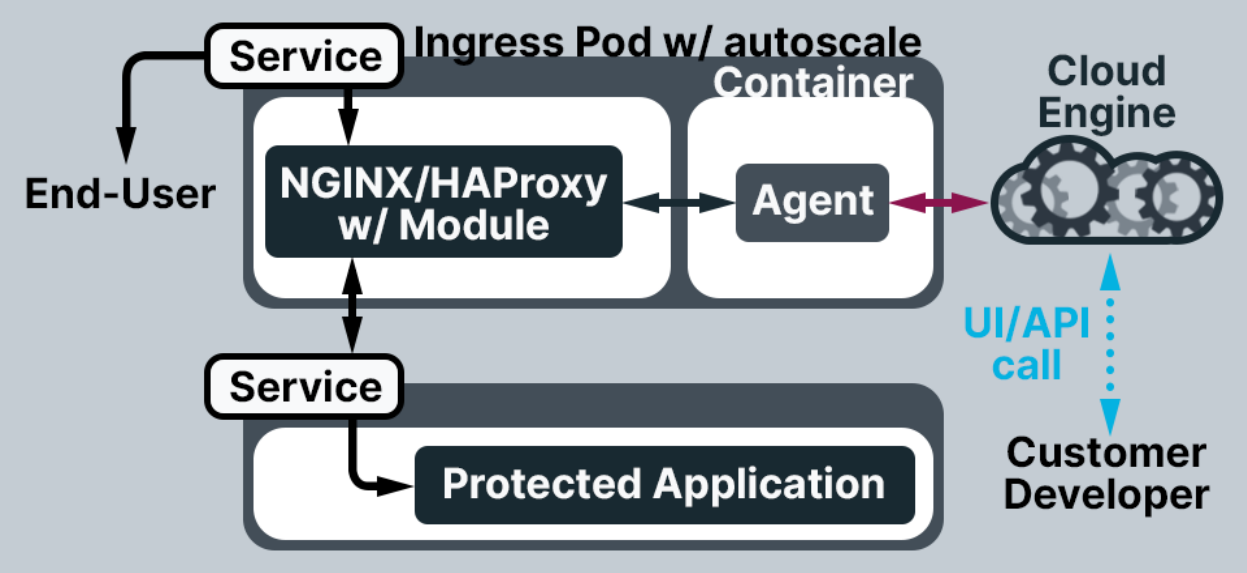
Single Pod, Multi-Container, Agent in Reverse Proxy Mode.



Multi-Pod, Single Container, Agent in Reverse Proxy Mode.



Multi-Pod, Single Container, Proxy with Module and Agent.

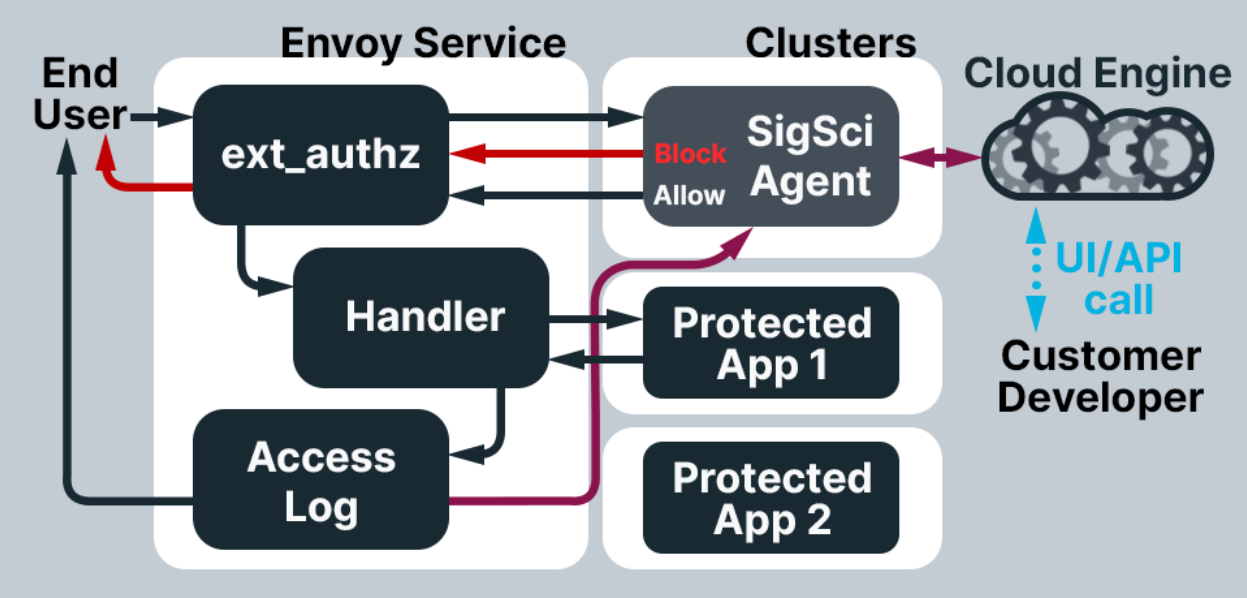


Multi-Pod, Multi-Container, Proxy with Module and Agent.

Relevant Links:

Lesson 5: The Envoy Deployment

Installing Next-Gen WAF into the Envoy environment also shows how adaptable our solution can be.



In Envoy, the ext_authz (External Authorization Filter) acts as a SigSci Module would, allowing or blocking incoming requests. The ext_authz is configured to check with our SigSci Agent, set up in a Envoy Cluster. The Agent decides which requests to allow/block and passes those directives back to the ext_authz. Any requests allowed through will be able to move to the Handler and be directed to the appropriate protected Application. As the response passes through the Access Log on the way downstream back to the end-user, response metadata is passed to the Agent for collection into the Cloud Engine.

Relevant Links:

Lesson 6: Supported Platforms

Next-Gen WAFs Core Deployment options cover a wide range of platforms, listed below. When you and your team are ready to select the option right for you, our Docs page has step install procedures for every option.

- NGINX
- Apache
- Java
- Ubuntu
- Red Hat
- Debian
- Amazon Linux
- AWS Lambda
- Windows
- Alpine
- Php
- Istio
- Go
- Kong
- Node
- Python

Relevant Links:

Quiz

1. The three Platform Plan Tiers, which determine feature availability are…
    a. Essential
    b. Professional
    c. Premier
    d. Enterprise
2. The Cloud Engine…
    a. Receives request data from customers installed Agents.
    b. Surfaces request data and metrics via the console, api, and integrations.
    c. Propagate Signal Science defined rules and Customer defined rules to relevant Agents.
    d. Handles request traffic directly.
3. The three broad Deployment Options are…
    a. Cloud WAF

   b. Core
   c. Edge
   d. Distributed
4. Of the three board Deployment Options, which one has a large variety of specific implementations (diagrams).
   a. Cloud WAF
   b. Core
   c. Edge
   d. Distributed

Technical Sign-Offs: You made it to the end! IF the proceeding doc met your satisfaction, please add your name below with the date. IF not, leave a comment on the relevant section above. Thank you!

Name	Date